

Cohomological Dimension of groups, EG

Def. ($\text{cd } G \leq n$) TFAE

(1) $\exists$ projective (or free) resolution by $\mathbb{Z}G$ -modules

$$0 \rightarrow P_n \rightarrow \dots \rightarrow P_0 \rightarrow \mathbb{Z} \rightarrow 0$$

(2) $H^i(G, M) = 0 \quad \forall \mathbb{Z}G\text{-module } M, \forall i > n$

(3) $H^{n+1}(G, M) = 0 \quad \forall \mathbb{Z}G\text{-module } M$

(4) $\forall$ proj. resolution

$$P_{n+1} \rightarrow \dots \rightarrow P_0 \rightarrow \mathbb{Z} \rightarrow 0$$

the kernel of $P_{n+1} \rightarrow P_n$ is projective,

pt. (Brown, Cohomology of groups, p. 184)

$1 \rightarrow 2 \rightarrow 3$ & $4 \rightarrow 1$ are obvious. Proj. ~~and~~ free is the Hilbert ^{syzygy} lemma

3 $\rightarrow$ 4 Complete + a projective resolution

$$\dots \rightarrow P_{n+1} \rightarrow P_n \rightarrow P_{n-1} \rightarrow \dots \rightarrow P_0 \rightarrow \mathbb{Z} \rightarrow 0$$

where $K \subseteq \text{image}(P_{nn} \rightarrow P_n)$, $L \subseteq \text{image}(P_{n+2} \rightarrow P_{n+1})$.
 We need to show K is projective.

We have $0 \rightarrow L \rightarrow P_n \rightarrow K \rightarrow 0$ (*)

We'll show this splits, so $P_n \cong K \oplus L$, so K is proj.

$\forall \mathbb{Z}[G]$ module N , an $(n+1)$ -cochain φ in $\text{Hom}(P_{n+1}, N)$

is a map $P_{n+1} \rightarrow N$ whose composition with $P_{n+2} \rightarrow P_{n+1}$ is 0, so φ can be viewed as a map $\varphi: L \rightarrow N$.

This is coboundary $\Leftrightarrow \varphi$ extends to $P_n \rightarrow N$. We are assuming every φ extends. In particular, $\text{id}: L \rightarrow L$ extends, so (*) splits. $\square$

Thm (Eilenberg-Ganea) If $n = \text{cd}(G) \geq 3$, then G is contractible n -complex freely, i.e. $\text{gd}(G) = n$.

Pf. Start building $K(G, 1)$ by attaching cells to kill homotopy groups. Get to Y $(n-2)$ -connected $(n-1)$ -dim free G -complex. This gives a

partial resolution

$$C_{n-1} \rightarrow C_{n-2} \rightarrow \dots \rightarrow C_0 \rightarrow \mathbb{Z} \rightarrow 0$$

and the kernel is $H_{n-1}(Y)$, thus

projective. Attach a collection of $(n-1)$ -spheres

st. $H_{n-1}(Y)$ is free. But $\pi_{n-1}(Y) \cong H_{n-1}(Y)$

when $n \geq 3$, so now attach n -cells to elements

in a basis to produce a contractible

$C - n$ -complex

□.

Stallings '68, Swan '69

$cd G = 1 \Rightarrow G$ is free, i.e. $gd G = 1$.

Eilenberg - Ganea Conjecture (EG)

$cd G = 2 \Rightarrow gd G = 2$

Whitehead Conjecture '41, asked as a question

(WH) Any subcomplex K of a ~~any~~ aspherical
2-complex L is aspherical,

Motivation: Quickly implies that knot complements
in S^3 are aspherical (since meridional disk is contractible)

~~True~~ True for ~~locally CAT(0)~~ locally CAT(0) complexes,
but generalizations (angle conditions. —)

(Frank)
J.F. Adams (1955) $\exists$ regular covering $\tilde{K} \rightarrow K$
which is a cyclic.

Jim Howie '83 If (WH) is false, $\exists$ counterexample

of one of the following forms:

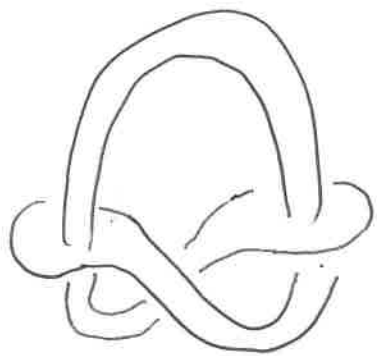
(a) L finite & contractible, $K = L$ — one 2-cell.

(b) $K = K_0 \subsetneq K_1 \subsetneq \dots$, $L = \bigcup K_i$ contractible,
each K_i finite, nonaspherical.

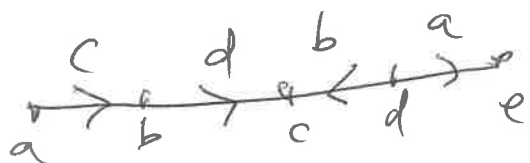
last '96: In fact (b) only is needed.

Howse '83 L finite 2-cx, AC equivalent to $*$.

$K = L - \text{one 2-cell}$. Then $K \simeq$ ribbon disk complement in B^4 .



These have 2-d spines analogous to Wirtinger presentation, given by LOT complexes
 $\hookrightarrow$ "labeled oriented tree"



$$ac = cb, bd = dc, db = bc, da = ae$$

Ribbon disk Conjecture

(and LOTs) are

Ribbon disk complements

aspherical.

Andrews - Curtis (1965)

$L = e^0 \cup e^1 \cup \dots \cup e^n \cup e^2 \cup \dots \cup e_n^2$
contractible (ie. a "balanced presentation" of 1).

Then L can be transformed to a point by
sliding 2-cells over each other, wedging
2-cells and unwedging 2-cells

(ie : invert generators
replace relator by a conjugate or inverse
replace r_i by $r_i r_j$)

stabilization suggested by reference $\rightarrow$ introduce new gen. y & relator y , and inverse

Topological application: If F is a homotopy 3-ball
then its double embeds in $\mathbb{R}^4$, so Poincaré
is false, counterexample exists in $\mathbb{R}^4$.

Expected to be false. See Bridson's work:
small complexes that require a huge # of moves,

The Kervaire (-Laudenbach) conjecture

(KL) If $G \neq 1$ is a group then the group
 $\hat{G} = \langle G, t_1, \dots, t_n \mid r_1, \dots, r_n \rangle = G * F_n / \langle\langle r_i \rangle\rangle$
is also nontrivial.

$M = n \times n$ matrix of powers of t_i in r_j .

If $\det M \neq 1$, then $\hat{G}$ maps onto $\mathbb{Z} / (\det M)\mathbb{Z}$.

(KL⁺) If $\det M \neq 0$ then $G \rightarrow \hat{G}$ is injective.

Top version we'll need:

K acyclic, $\pi_1 K \neq 1$, $L = K \cup (n \text{ 1-cells}) \cup (n \text{ 2-cells})$
also acyclic.

Then $\pi_1 K \rightarrow \pi_1 L$ is not trivial.

Also, comes up in ~~math~~ studying cobordisms,

EX, Mazur's contractible 4-mfld cannot be
built without 1-handles.

Gerstenhaber - Rothaus '82

If H is a compact connected Lie group, $\det H \neq 0$
then every $G \rightarrow H$ extends to $\tilde{G}$.



So residually finite groups satisfy (KL^+) .

~~Howe~~ Howe '82

Locally indicable groups
satisfy (KL^+) .

Klyachko '92, L. Chen '26

G torsion-free satisfies (KL^+) when $n=1$.

- other work, e.g. Klyachko then extending G -l.

Thm (B-Brady, '97)

At least one of (WH) or (EG) is false.

Outline P = Poincaré homology 3-sphere

$$= \widetilde{SO(3)} / \tilde{I}, \text{ where } I \subset SO(3) \text{ is } A_5, \text{ o.p. symmetries of a dodecahedron}$$

= dodecahedron with opposite faces identified by the smallest positive twist.

K = spine of I -point, 2-dim acyclic complex,

$$\pi_1 K = \pi_1(S^3 - 120 \text{ pts})$$

$$\pi_1 K = \tilde{I} \text{ order } 120$$

$$\pi_2 K = \pi_2(S^3 - 120 \text{ pts}) = \mathbb{Z}^{119}$$

Choose a flag triangulation of K , let G = RAAG on K .

$\phi: G \rightarrow \mathbb{Z}$ sends all generators to 1.

X = CAT(0) cube cx for G (Solovetti cx)

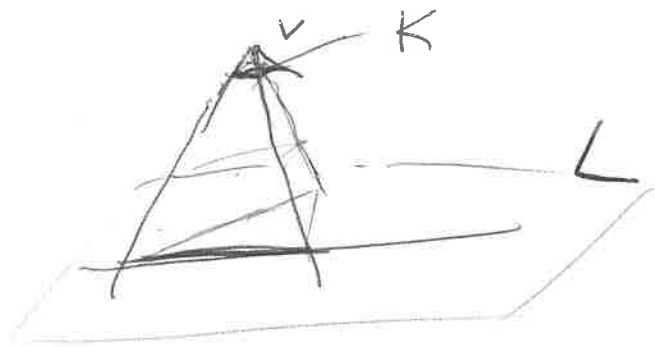
$f: X \rightarrow \mathbb{R}$ ϕ -equiv. map extending $\phi: X^{(0)} \rightarrow \mathbb{Z} = \mathbb{R}^{(0)}$ in a "linear" fashion.

The descending and ascending links $L_K^-(v), L_K^+(v)$ at any vertex are copies of K , so acyclic.

More exact $\Rightarrow$ level sets L_f of f are acyclic.

$H = \ker(\phi)$ acts freely & cocompactly on level sets, $(H \cong \mathbb{Z}^2)$ so $\text{cd} H = 2$

Geometric input level sets L contain arbitrarily large copies of K .



Now suppose $H \curvearrowright W$ 2-d contractible complex

Constant H -equiv ^{free simplicial} map $L \rightarrow W$,

L is H -cocompact $\Rightarrow$ point inverses are uniformly bounded

Then the composition $K \xrightarrow{\text{"large inclusion"}} L \rightarrow W$

has ϵ -small point inverses. If Y is the image,

then $K \xrightarrow{\epsilon} Y$ has an approximate left inverse,

so $\pi_2(Y) \neq 0$, contradicting (WH).

$\square$.

Addendum If $Y \simeq K \cup \underbrace{1\text{-cells} \cup 2\text{-cells}}_{\text{finitely many}}$

then there is a contradiction w/o (WH), so (EG) is false.

Pf. Y is a finite subcomplex of $W = \text{contractible } 2\text{-CX}$,
so $Y \subset Z$ finite subex of W , $Y \hookrightarrow Z$ trivial in π_1 .

Now ~~$H_2(Z, \mathbb{Z}) \neq 0$~~

$H_2(Z) = 0$ and $H_1(Z)$ is free ab. (otherwise $H_2(Z, \mathbb{Z}/p) \neq 0$ for some p)

so $\hat{Z} = Z \cup (2\text{-cells})$ is acyclic.

so $K \hookrightarrow \hat{Z} \simeq K \cup \underbrace{1\text{-cells} \cup 2\text{-cells}}_{\text{same \# since both acyclic}}$

But $\pi_1 K$ is finite, so by GR satisfies (KL^+) ,
so contradiction. $\square$.

Conjecture K finite simplicial 2-CX. Then $\exists \varepsilon > 0$
s.t. if $K \rightarrow Y$ surjective PL-map, then
 $Y \simeq K \cup 1\text{-cells} \cup 2\text{-cells}$.

Brady-Leary - Munk's 116

F group that acts on an acyclic 2 -complex
with finite stabilizers, and with $\text{Fix}(F)$ acyclic
 $\nexists$ finite $F < G$, but doesn't act in this
way on any contractible complex with contractible
fixed sets.